Maximo Monitor 9.1 Managed Gateway Modbus Lab

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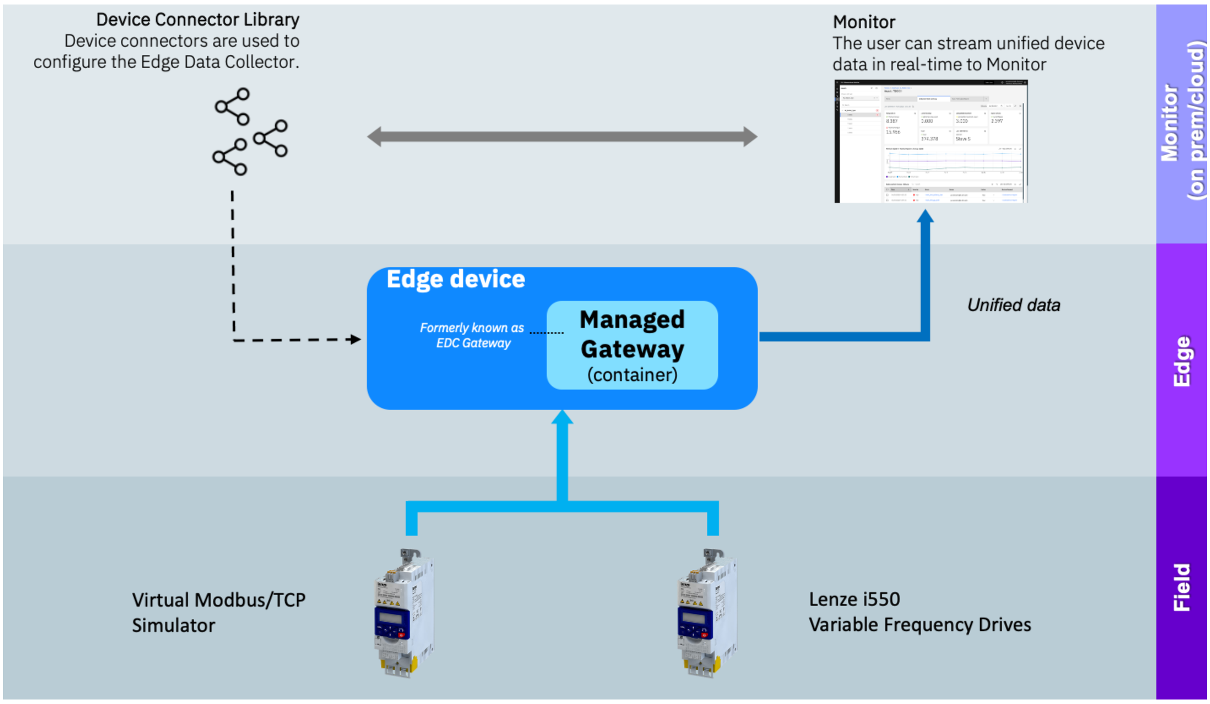
Overview

In this lab you will learn how to setup a Managed Gateway within Maximo Monitor 9.1 and adding a couple of OT devices from the Device Library.

You will learn the steps needed to successfully setup a Managed Gateway using some virtual simulated Variable Frequency Drives (VFD) that will deliver the device data over Modbus/TCP via the Managed Gateway into Maximo Monitor using the pre-configured connector for Lenze i550 VFD from the Device Connector Library.



1. In the provided URL, replace **admin** with **api**, then press **Enter**.
2. An error message will appear — this is expected.
3. Next, replace **api** back with **admin** in the URL and press **Enter** again.

Note: When switching between the MAS module and MAS Monitor, you may be prompted to proceed to a site without a CA-authority-signed certificate. Accept the warning to continue.

The exercises will cover:

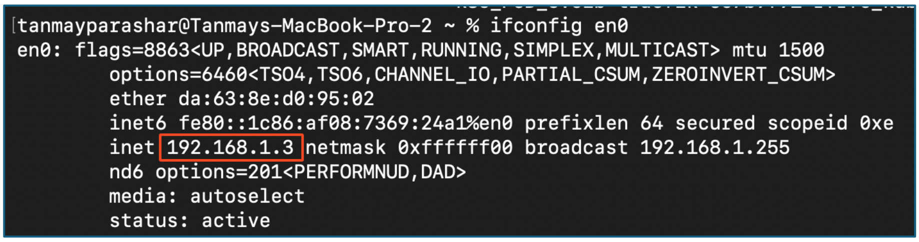
- Setting up the simulator environment
- Create a Managed Gateway
- Add devices to the Manage Gateway
- Verify data flow from Modbus/TCP Variable Frequency Drives all the way into Maximo Monitor

Exercise 1. Setup Simulator Environment

In this Exercise you will learn how to: Install the Modbus simulator

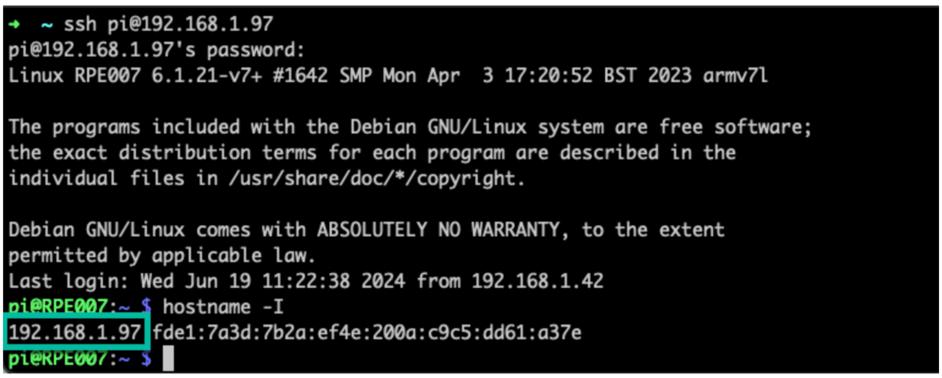
1. You need the IP address of the machine where you run the simulator no matter which version you are using. You will need that later when configuring the Managed Gateway.

- **Get IP Adddress on macOS**
  Use the following command: **ifconfig en0**



```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether da:63:8e:d0:95:02
        inet6 fe80::1c86:af08:7369:24a1%en0 prefixlen 64 secured scopeid 0xe
        inet 192.168.1.3 netmask 0xffffff00 broadcast 192.168.1.255
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
```

- **Get IP Address on LINUX**
  Use the following command: **hostname -I**

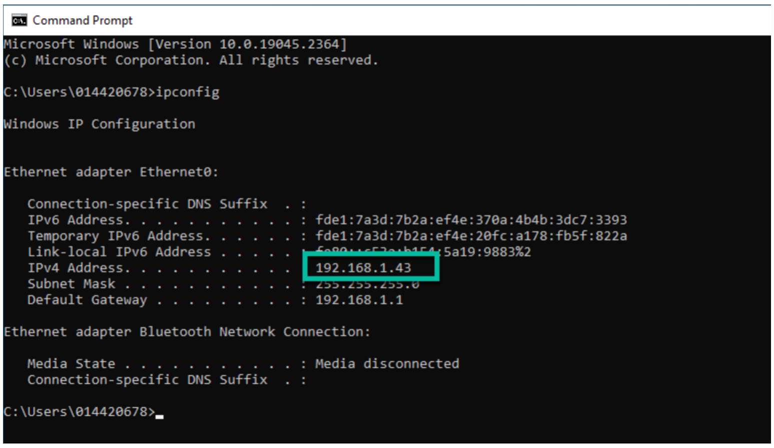


```
→ ~ ssh pi@192.168.1.97
pi@192.168.1.97's password:
Linux RPE007 6.1.21-v7+ #1642 SMP Mon Apr  3 17:20:52 BST 2023 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jun 19 11:22:38 2024 from 192.168.1.42
pi@RPE007:~ $ hostname -I
192.168.1.97 fde1:7a3d:7b2a:ef4e:200a:c9c5:dd61:a37e
pi@RPE007:~ $
```

- **Get IP Address on Windows**
  Use the following command: **ipconfig**

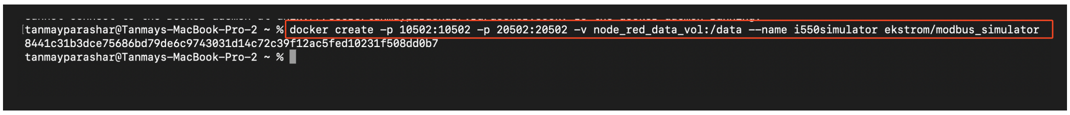


2. **Install Docker:** There are different ways to install the Docker engine depending of the package and the OS. One multi-platform option is **Rancher Desktop.**

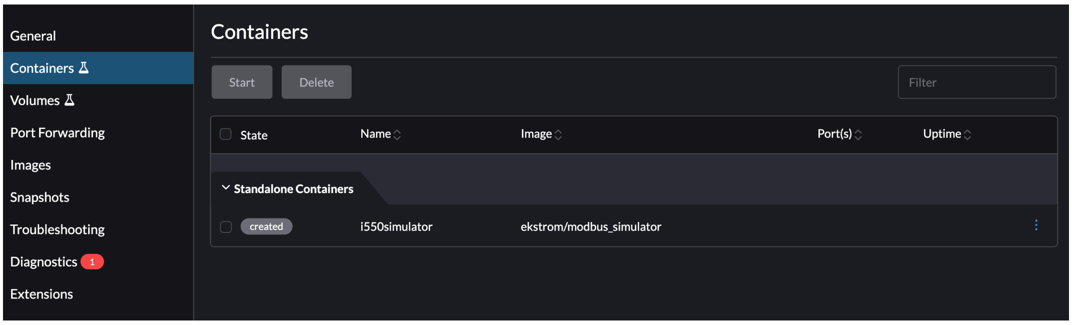
3. It is fairly easy to install Rancher Desktop, as you just have to follow this guide: [Running Docker locally]()

4. **Create the Docker container**: Open a terminal or command window and run the following command:

   docker create -p 10502:10502 -p 20502:20502 -v node_red_data_vol:/data --name i550simulator ekstrom/modbus_simulator



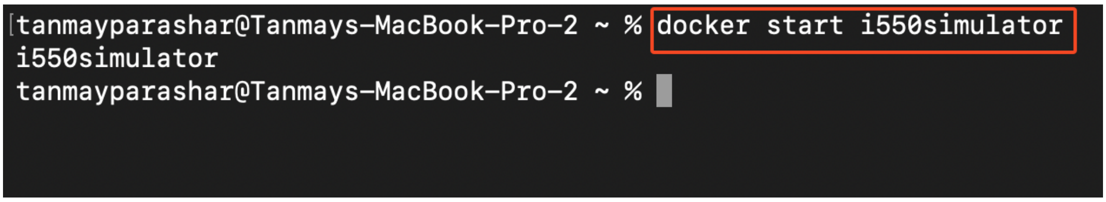
You can see a container got created in Rancher Desktop



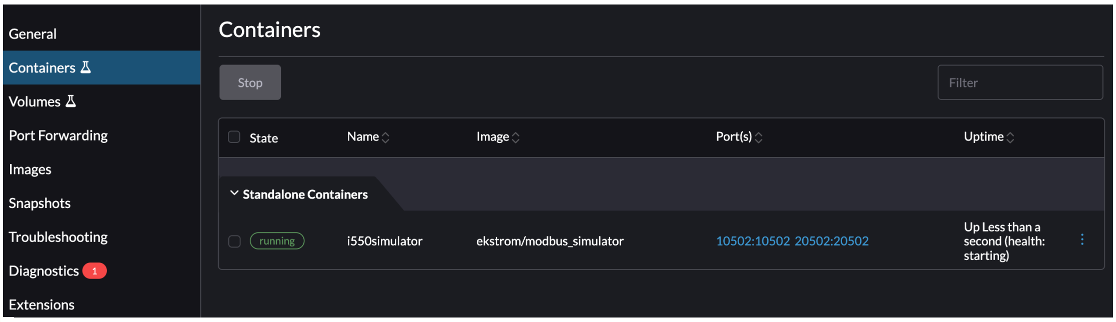
This command **sets up a Modbus simulator container** (i550simulator) that exposes ports 10502 and 20502 to the host and stores its persistent data in the node_red_data_vol volume. The container is only created.

5. **Start the Docker container:** Run the following command to start the container.

docker start i550simulator



You can see simulator is running in Rancher Desktop

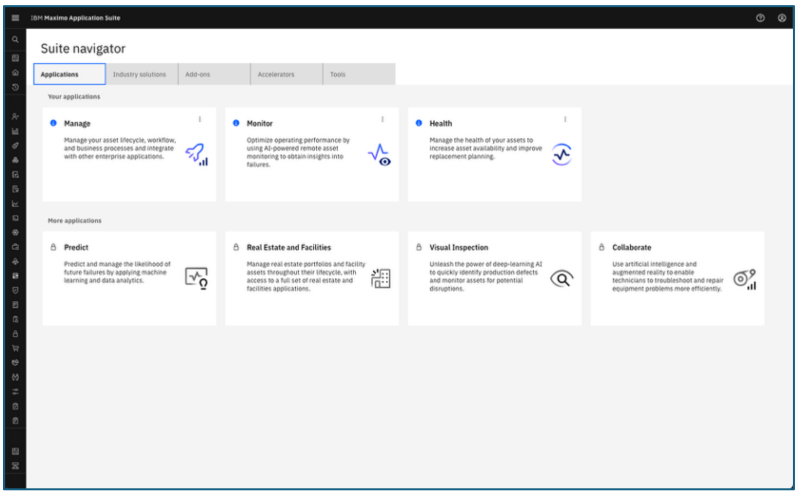


Note: The simulator is now active and the random and dynamic values will change every 30 second. It will run in the background and not produce any output in the terminal/command window.

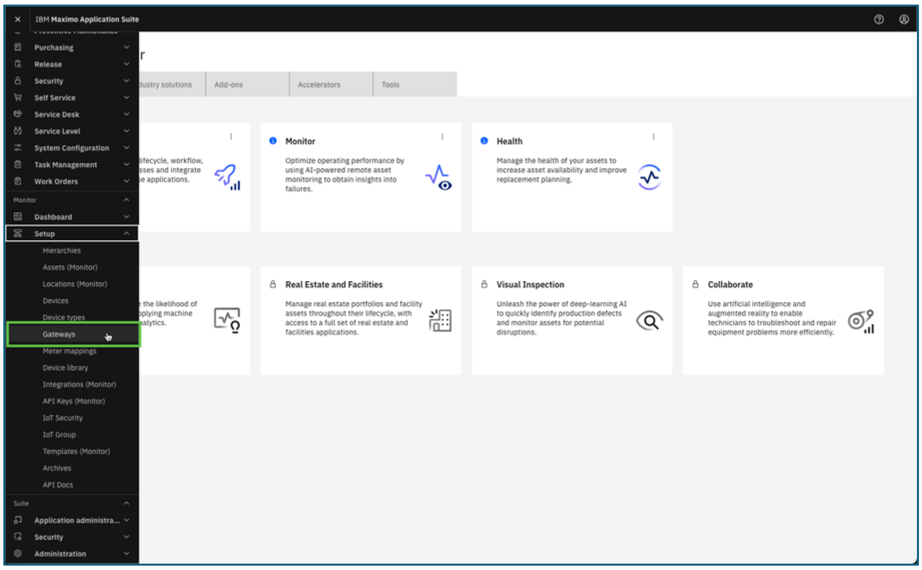
Exercise 2. Create a Managed Gateway

In this Exercise you will learn how to create the Managed Gateway in Monitor.

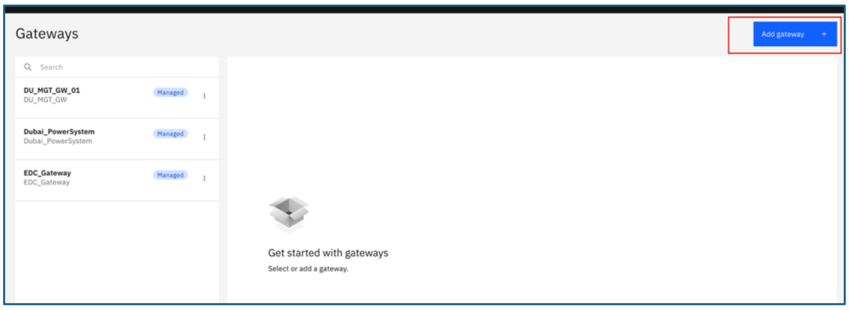
1. Login to Maximo Application Suite with your credentials.



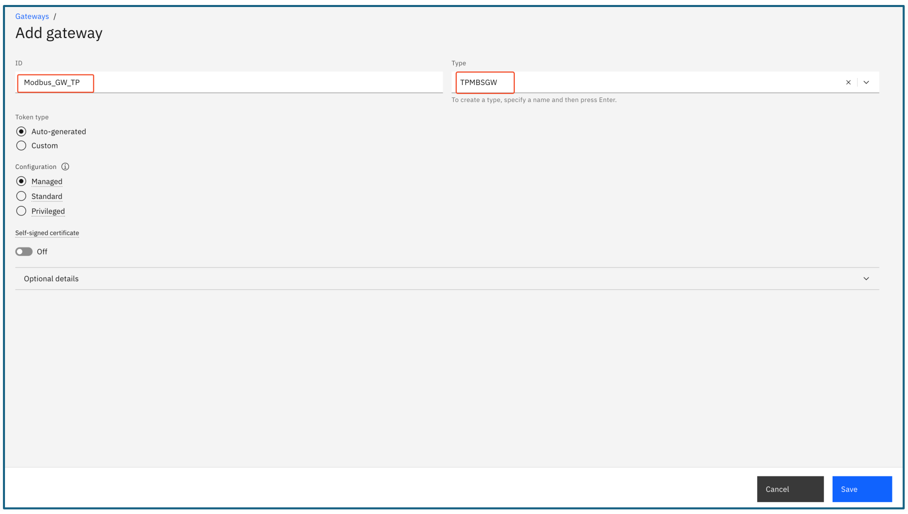
2. Expand Setup under the Monitor section in the left menu and select **Gateways**:



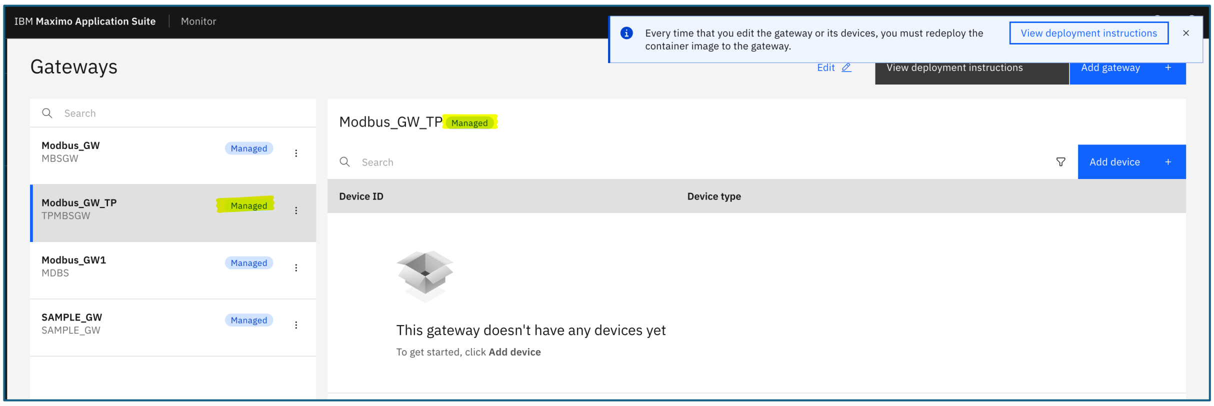
3. Select **Add Gateway**.



4. Provide gateway id and gateway type name. You can type preferred name and click enter.
    - ID- **Modbus_GW_XX** (Replace XX with your initials)
    - Type- **XXMBSGW** (Replace XX with your initials)
    - Token Type- **Auto-generated**
    - Configuration- **Managed**



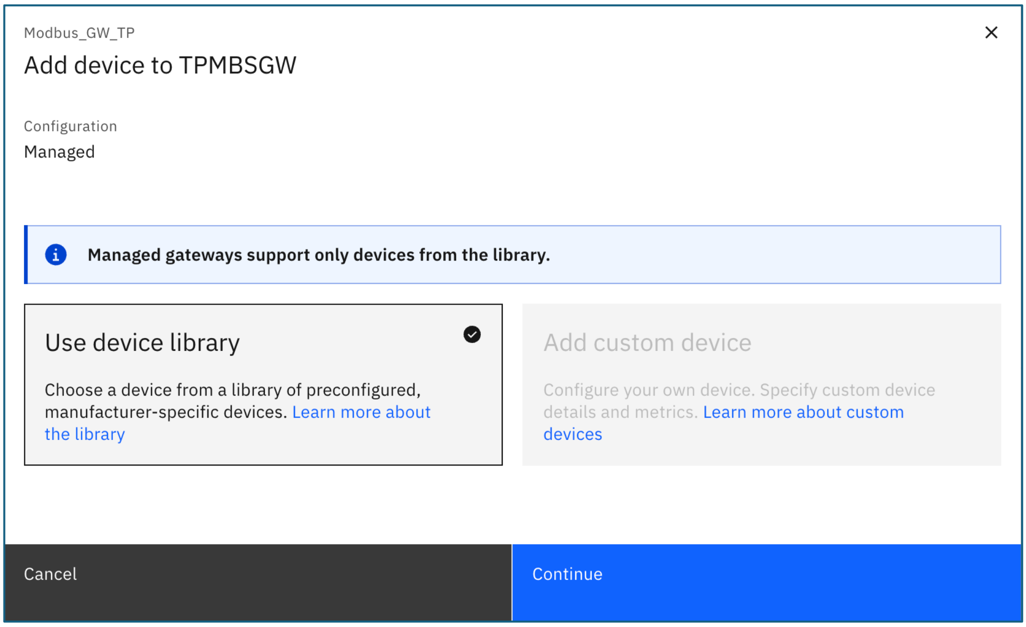
5. Click on **Save**.
6. You will now see your new Managed Gateway, including a **Managed** tag in both the list of Gateways as well as in the gateway definition.



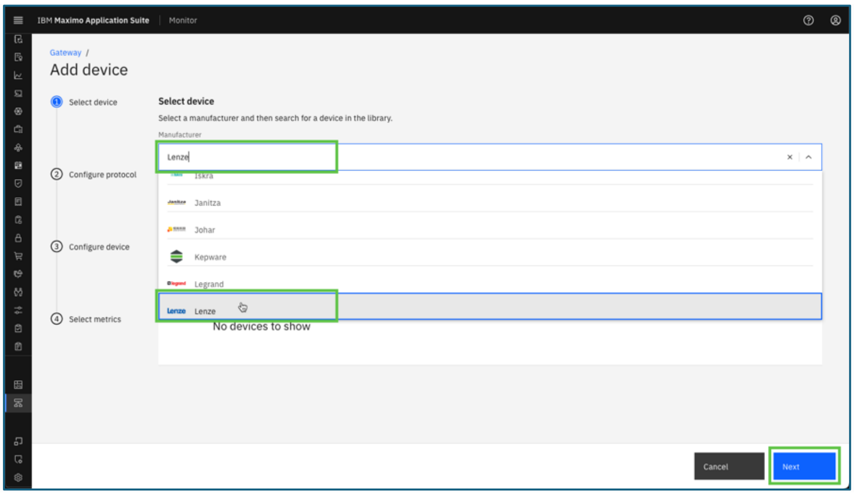
Exercise 3. Add an Industrial Device

In this Exercise you will learn how to add the first industrial device to the Managed Gateway.

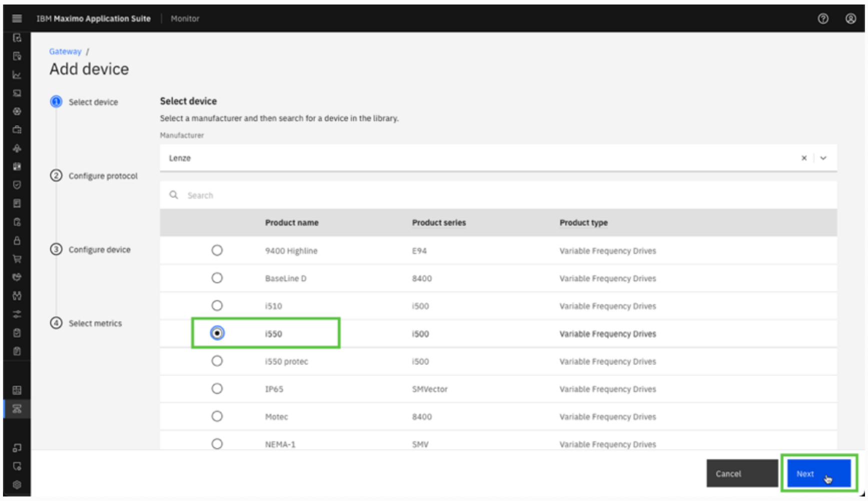
1. Click on **Add Device** and select **Use device library**. (Contd. From last exercise)



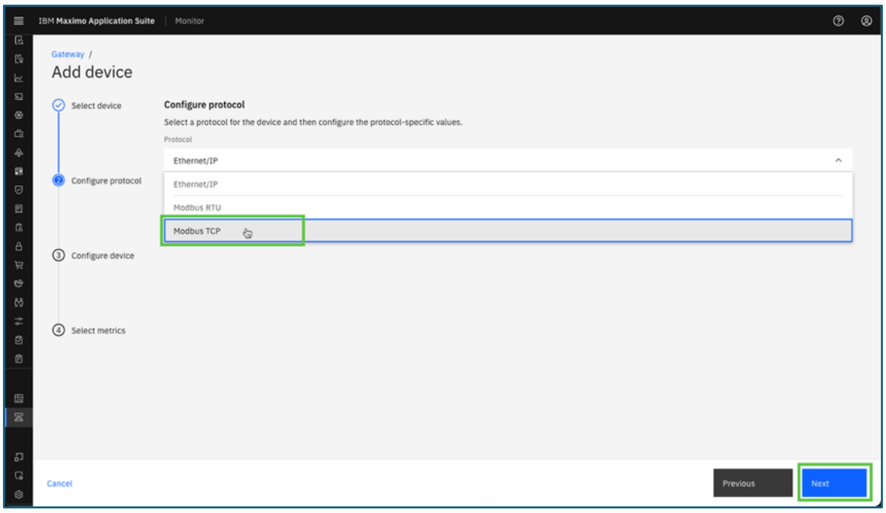
2. Search for **Lenze** in the manufacturer drop-down and select it. Click **Next**:



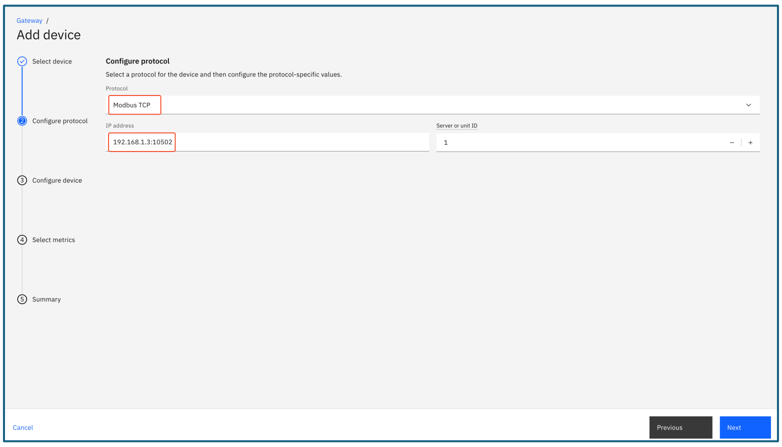
3. Select the **i550** product and click on **Next**.



4. Select the **Modbus TCP** protocol. Click on **Next**:

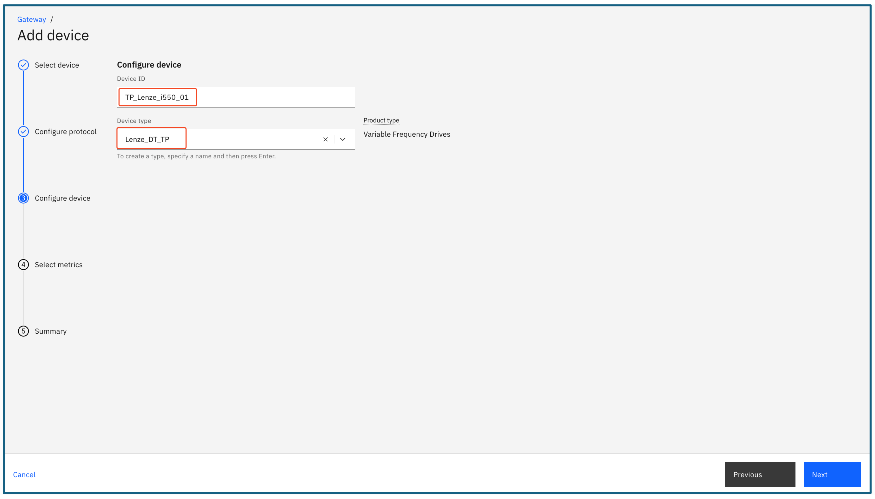


5. Now it is time to use the IP address of the simulator and combine it with port number 10502 separated by **:** , like **192.168.1.64:10502**. Click on **Next**.

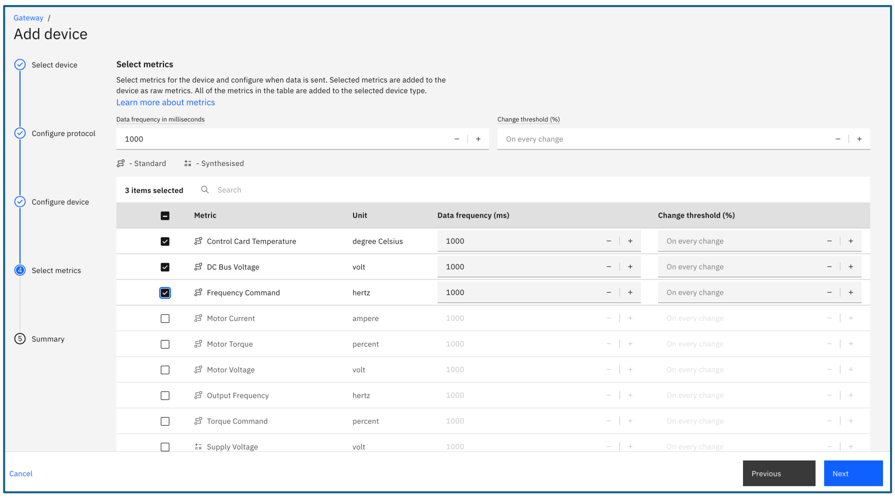


6. It will ask you to configure device and provide your own device id and device type. Click on **Next**.

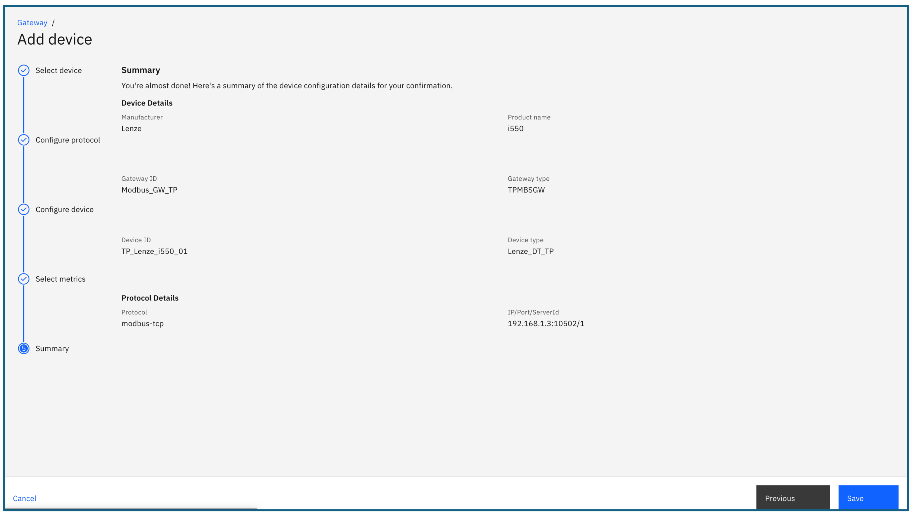
   - Device ID- **XX_Lenze_i550_01**
   - Device type- **Lenze_DT_XX** (Replace XX with your initials)



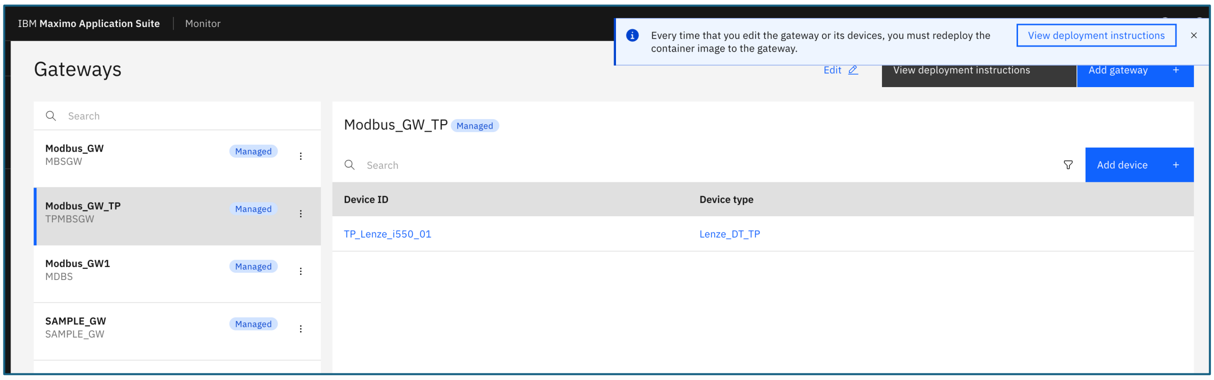
7. Select these three metrics and click on **Next**.



8. Click on **Next -> Save**.



9. You have now successfully added the first OT device to your Managed Gateway.

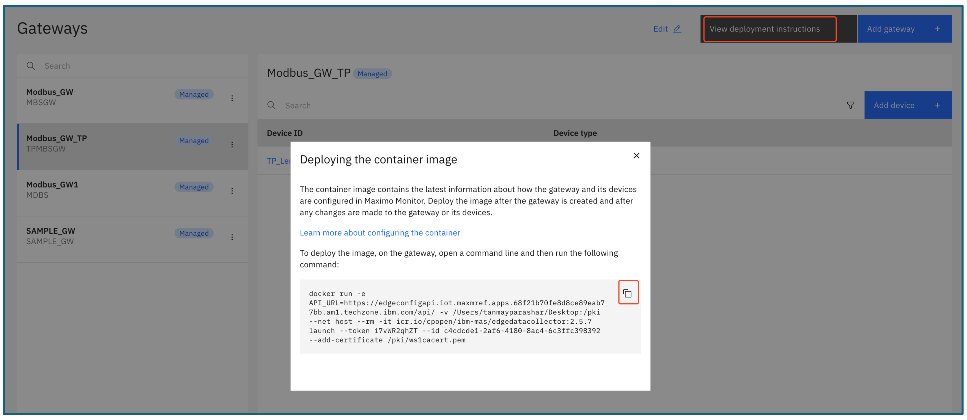


Exercise 4. Deploy and verify data inflow

In this Exercise you will learn how to:

- Deploy the Managed Gateway
- Verify data inflow

1. While looking at your Managed Gateway in the Gateways list, press the **View deployment instructions**. Click on the docker command to copy it to the clipboard.

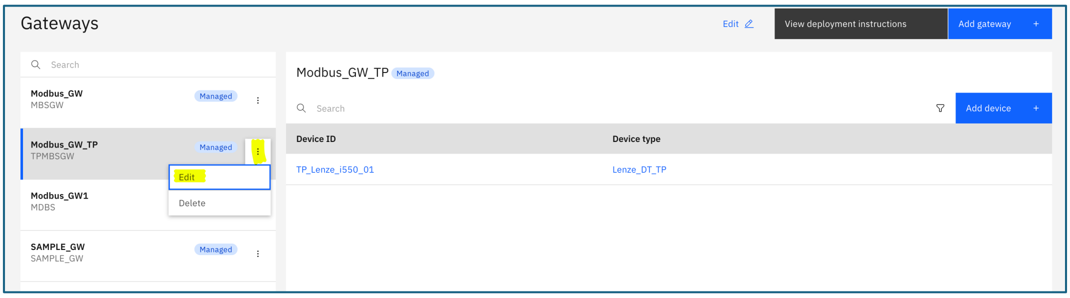


2. Once you run this docker command in Terminal/Command Prompt you will encounter the below error.

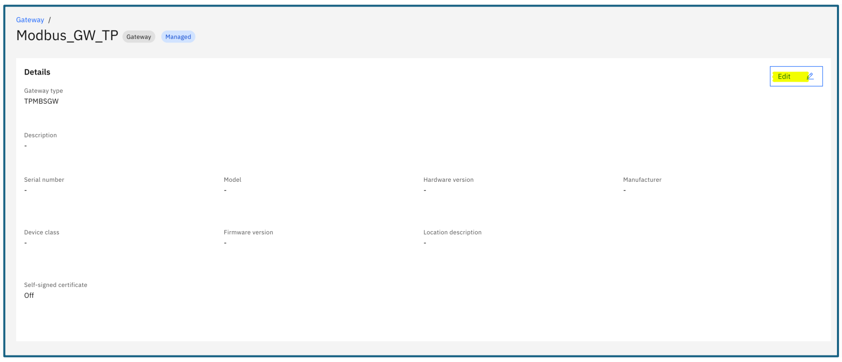
```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker run -e API_URL=https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/ --net host --rm -it icr.io/cpopen/ibm-mas/edgedatacollector:2.5.7 launch --token i7vWR2qhZT --id c4cdcde1-2af6-4180-8ac4-6c3ffc398392
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-26T18:24:07.208024Z: INFO: edge-cli/src/lib.rs:106: Arguments: launch --token i7vWR2qhZT --id c4cdcde1-2af6-4180-8ac4-6c3ffc398392
2025-10-26T18:24:07.211473Z: INFO: edge-cli/src/lib.rs:108: Version: 2.5.7
2025-10-26T18:24:07.211559Z: INFO: edge-cli/src/lib.rs:109: Git sha1: cab6ca6f2bf101743db0f093cd83bb9763bc4668
2025-10-26T18:24:07.211609Z: INFO: edge-cli/src/lib.rs:110: Branch: master
2025-10-26T18:24:07.211656Z: INFO: edge-cli/src/lib.rs:111: Build number: 748
2025-10-26T18:24:07.211700Z: INFO: edge-cli/src/lib.rs:112: Build ID: 41027330
2025-10-26T18:24:07.347811Z: INFO: /omnioiot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdcde1-2af6-4180-8ac4-6c3ffc398392/
error sending request for url (https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdcde1-2af6-4180-8ac4-6c3ffc398392/)

Caused by:
    0: client error (Connect)
    1: error:1416F086:SSL routines:tls_process_server_certificate:certificate verify failed:ssl/statem/statem_clnt.c:1915: (unable to get local issuer certificate)
    2: error:1416F086:SSL routines:tls_process_server_certificate:certificate verify failed:ssl/statem/statem_clnt.c:1915:
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %
```

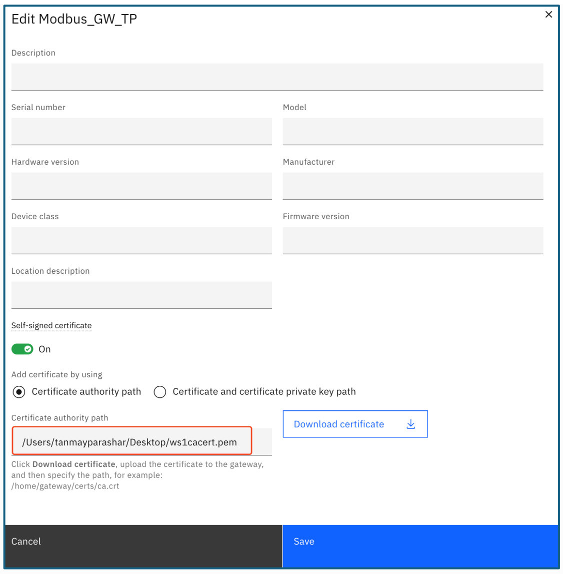
3. To fix the above error, go to Gateways you created in Monitor and click on Manage Gateway (three dots) and then click on **Edit**.



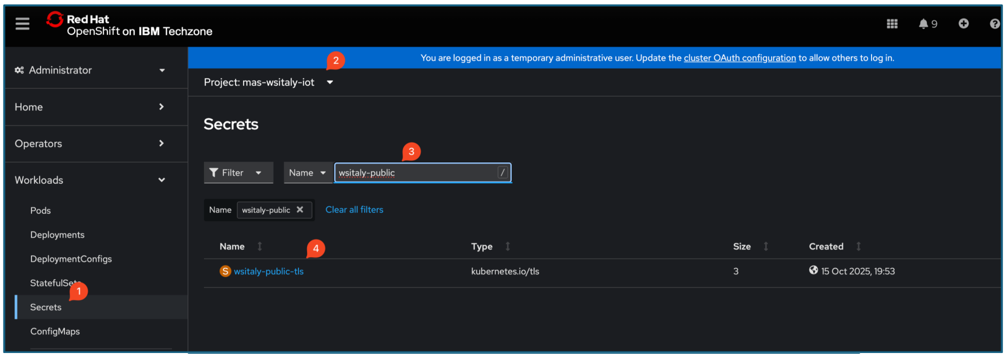
4. Click on **Edit**.

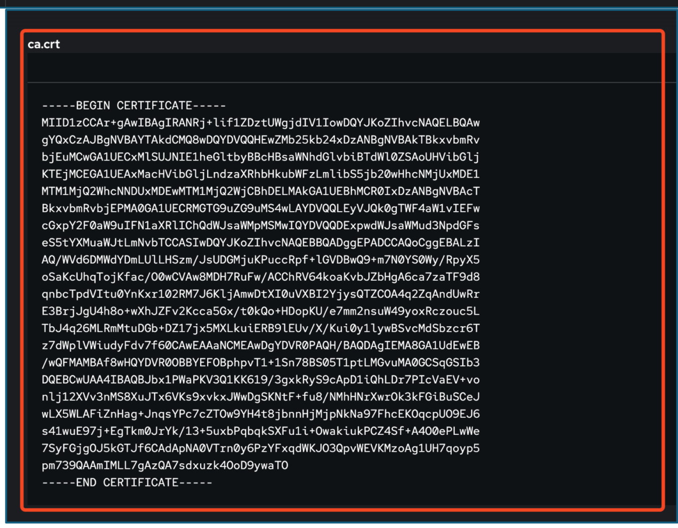


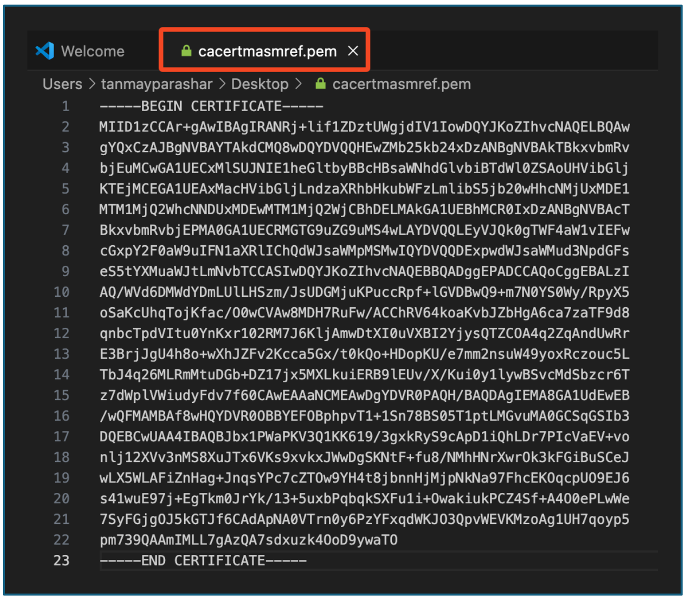
5. Enable **Self-signed certificate** and download the certificate from OCP and put the path of the certificate.



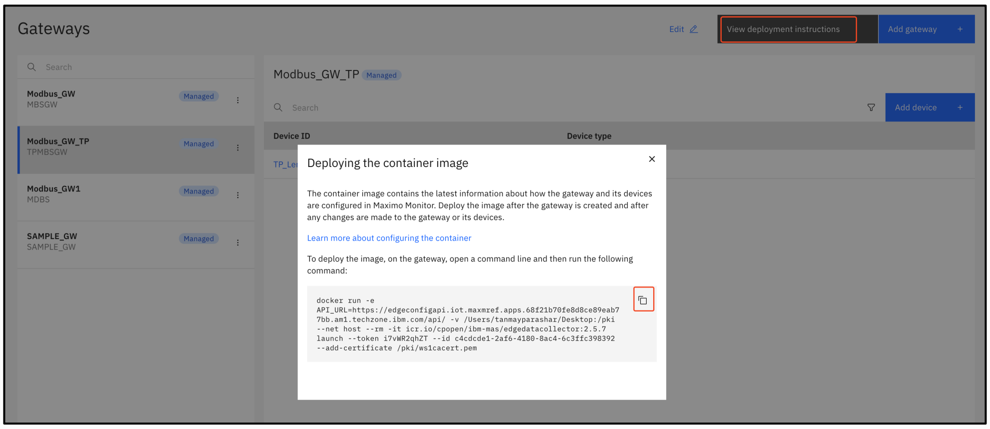
Note: To download the certificate from cluster navigate to **Workloads -> Secrets -> Project-> IoT -> Search public TLS ->** Download **ca.cert** and save the certificate with **.pem** extension.



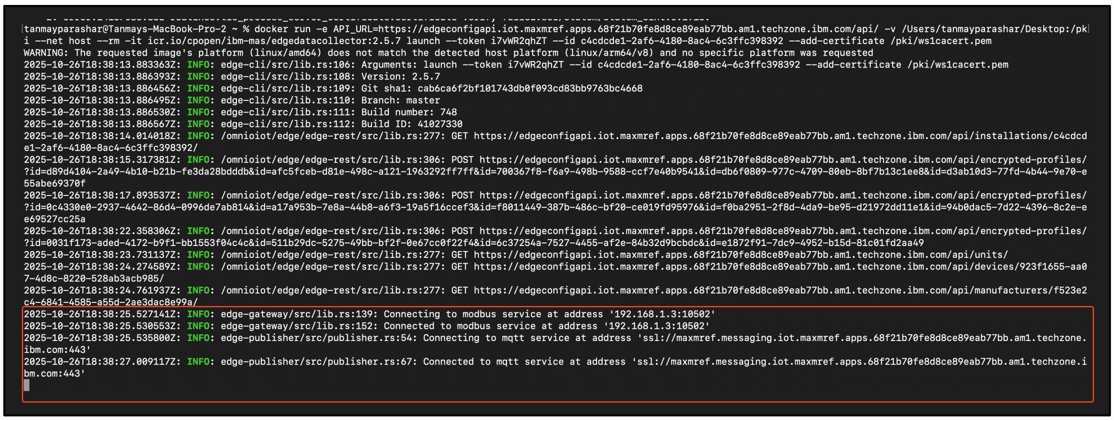




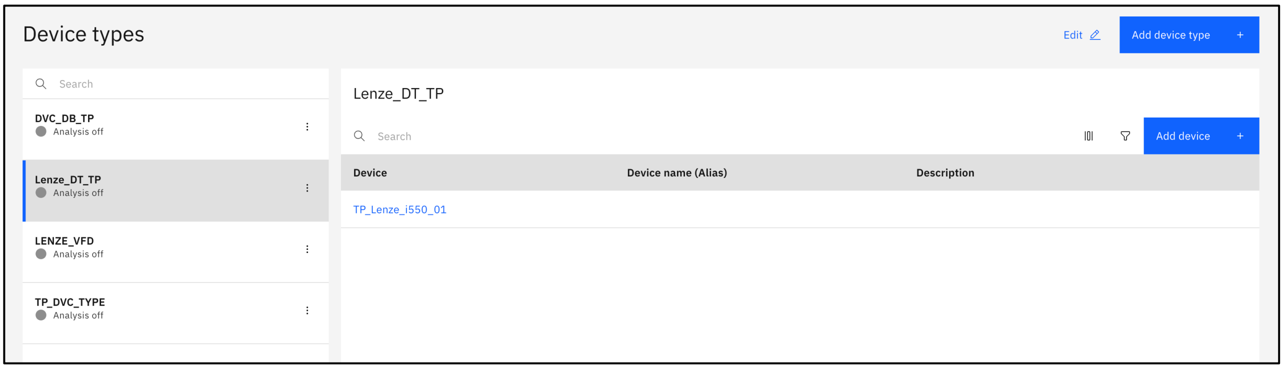
6. Click on **Save** and go to **Gateways** and copy the docker command by clicking on View deployment instructions.



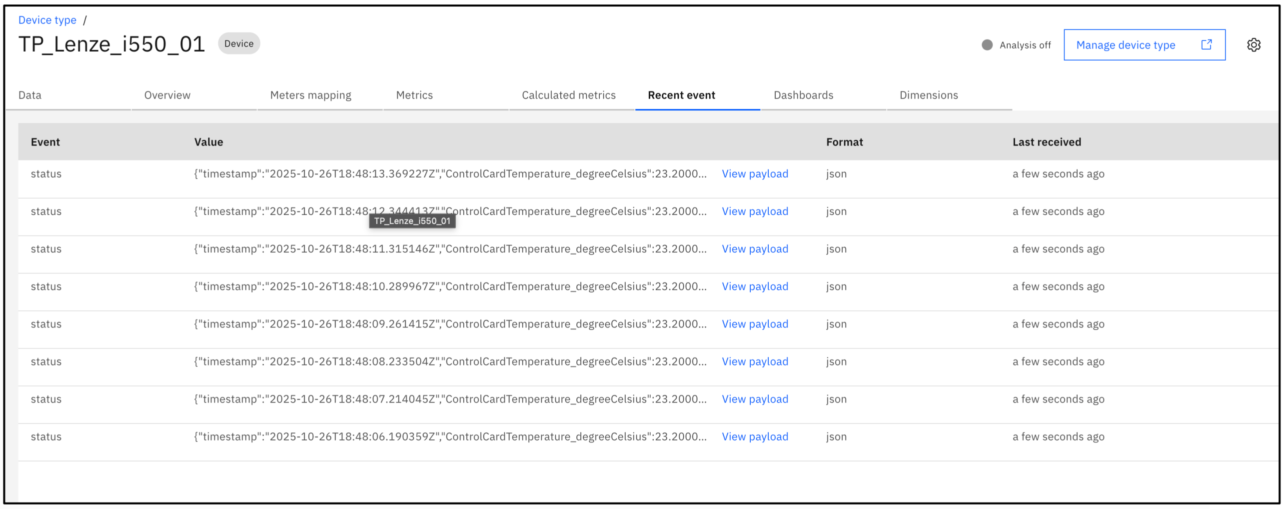
7. Open a terminal window (Mac/Linux) or Command window (Windows) where you want to run the Managed Gateway and then paste the docker command line from the clipboard. Click **enter** to execute it, and you should see something similar to the following.



8. Verify the selected Lenze data in Monitor. Click to open the **XX_Lenze_i550_01** device.



9. Navigate to **Recent event** and wait for a minute until the first message is coming through.



Note- If the recent event is not showing any payload. Then go to the terminal where you run the docker command and copy the MQTT service append https before the URL and launch in the browser like:

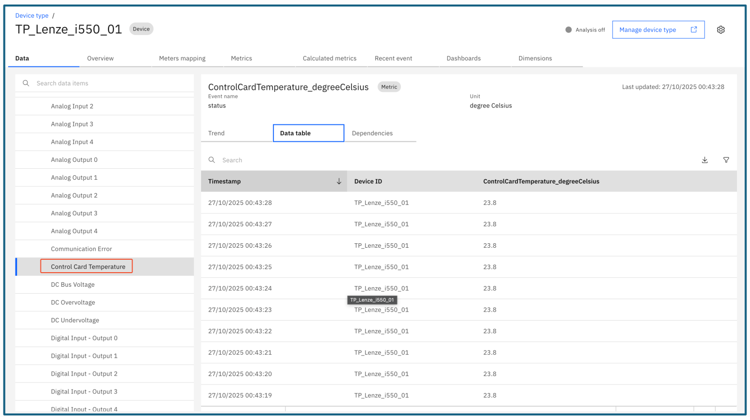
```
2025-10-26T18:38:25.535800Z: INFO: edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443'
2025-10-26T18:38:27.009117Z: INFO: edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443'
```

https://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443

and accept the certificate. Refresh the device you will see the payload.

10. Click on **View payload** and see the data points being send to the Event name status.

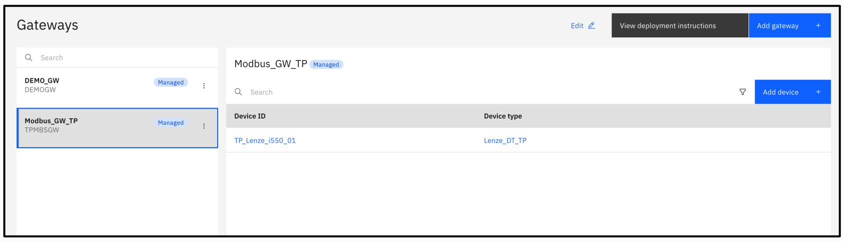
11. To see the data in the metrics. **Go to Data -> Control Card Temperature -> Data Table.**



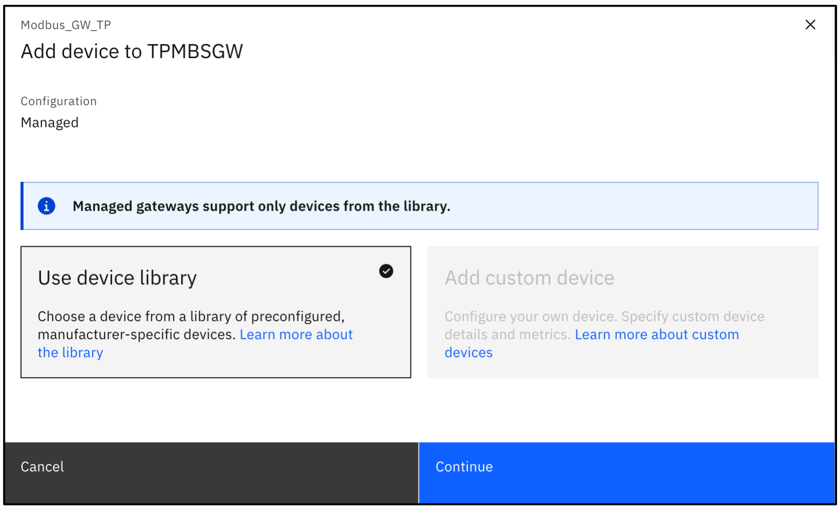
Exercise 5. Add another Industrial Device

In this Exercise you will learn how to add a second industrial device to the Managed Gateway.

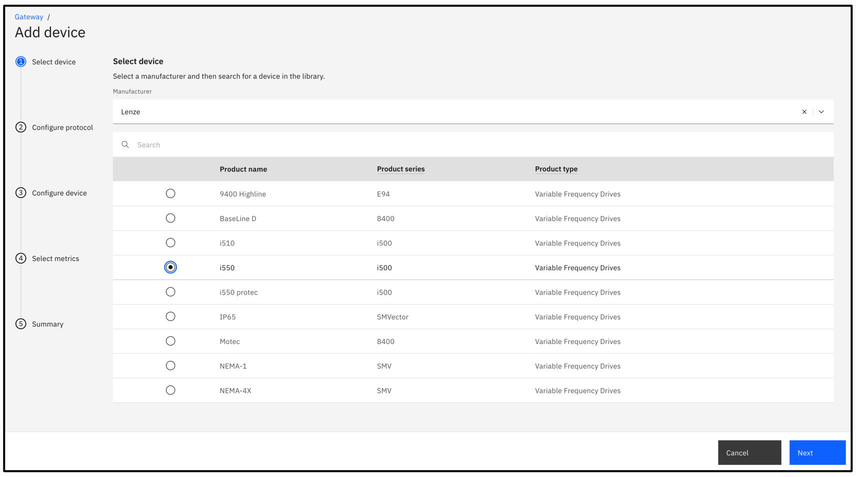
1.Open the Gateway in Monitor that you have created.



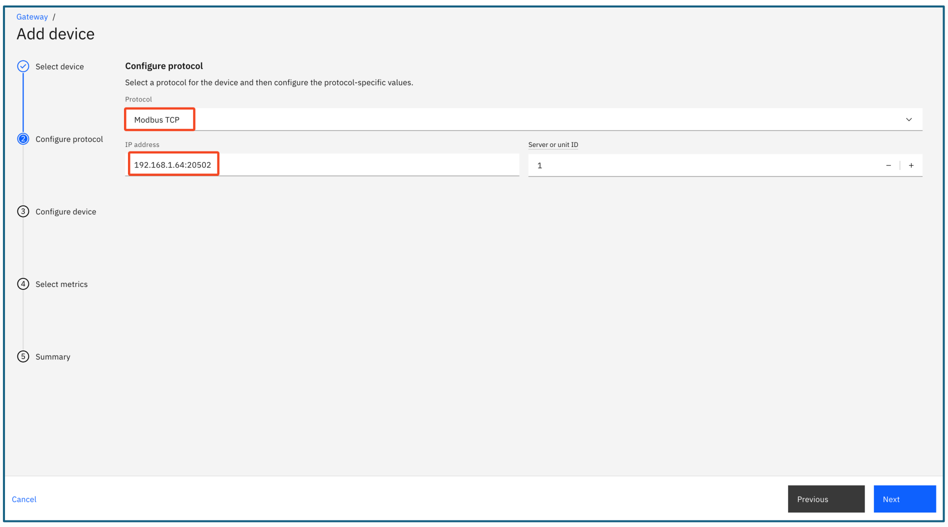
2. Click on **Add device**.



3. Select Manufacturer: **Lenze** and product: **i550** and click on Next.

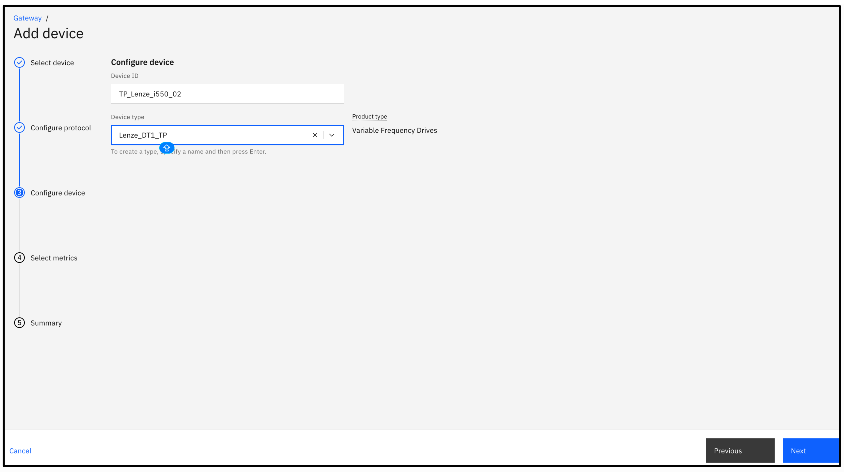


4. Now it is time to use the IP address of the simulator and combine it with port number 20502 separated by **:** , like **192.168.1.64:20502**. Click on **Next**.

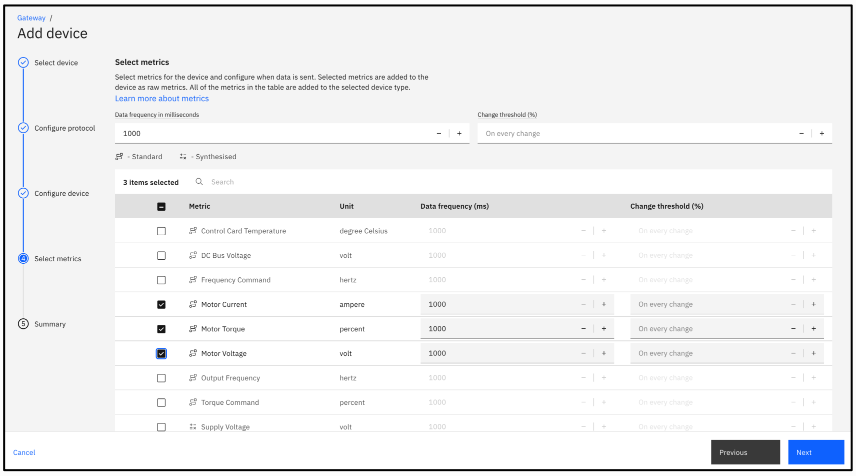


5. Enter the Device ID and Device Type:

- Device ID: **XX_Lenze_i550_02**
- Device type: **Lenze_DT1_XX**

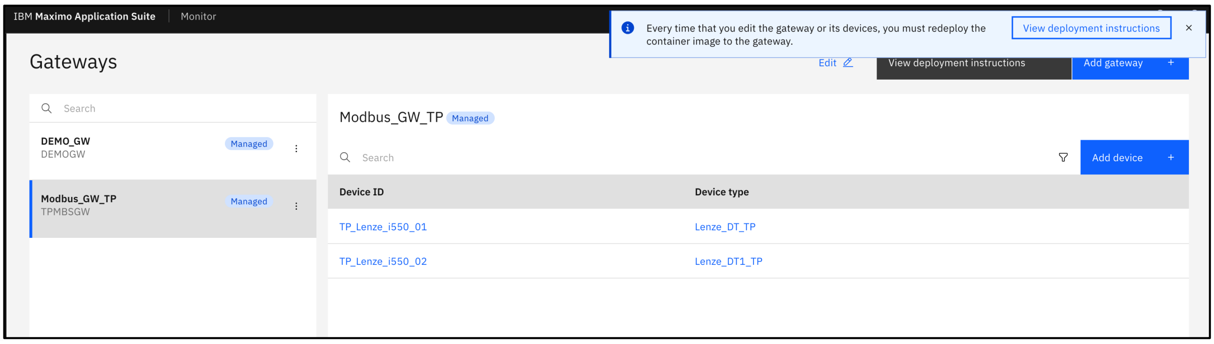


6.Select these three metrics and click on **Next**.



7.Review the Summary and click on **Save**.

8. Both the devices will be reflecting under the gateway.

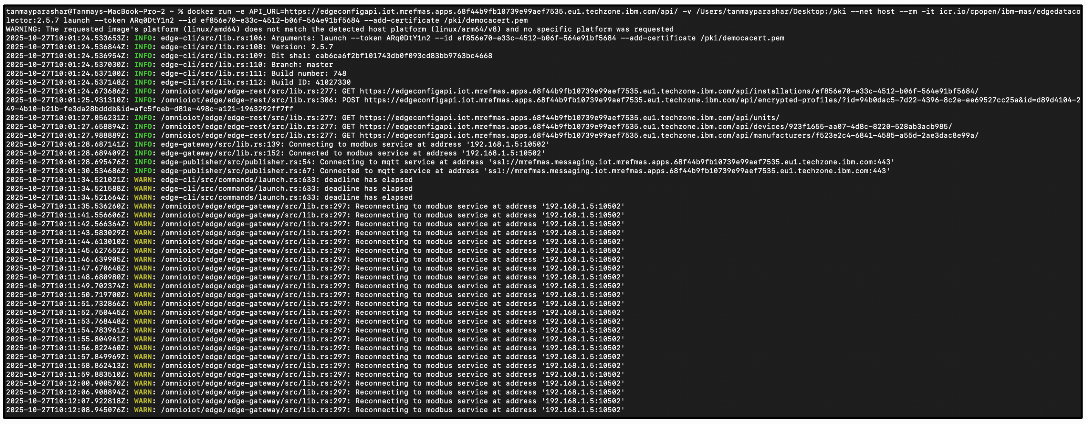


Exercise 6. Deploy and view data in dashboard

In this Exercise you will learn how to:

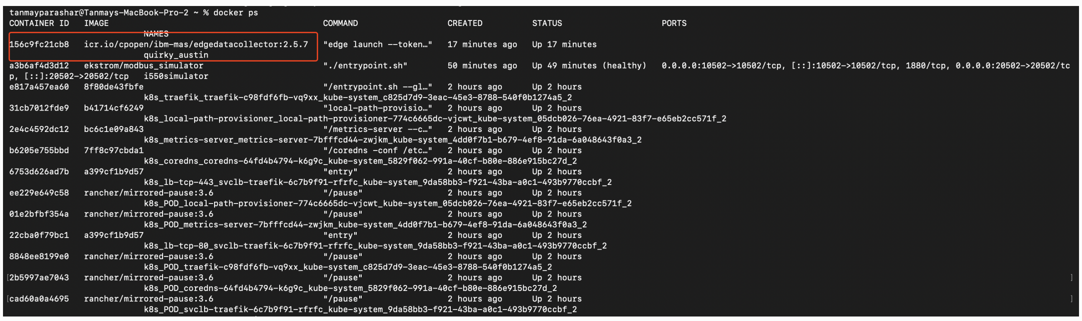
- Stop and re-deploy the Managed Gateway
- View the incoming data in Monitor dashboards

1.Go to the Terminal or Command Prompt where the current Managed Gateway docker container is running.



2.Press **control-p** and then **control-q** to get back to the prompt.

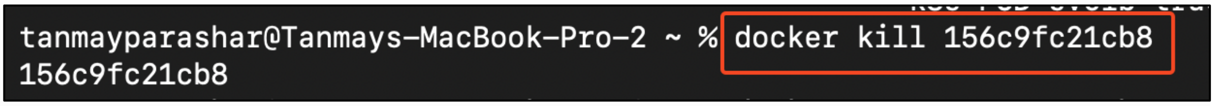
3.Then run the command **docker ps** to view the running docker containers.



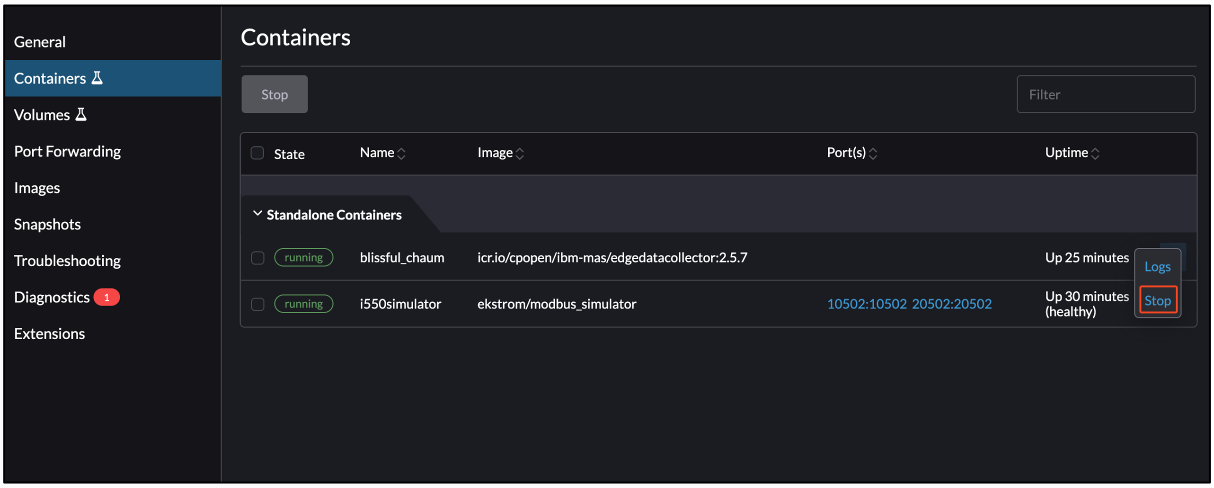
4. Copy the Container ID from the previous command i.e. 156c9fc21cb8.

And then run this command **docker kill <container id>**

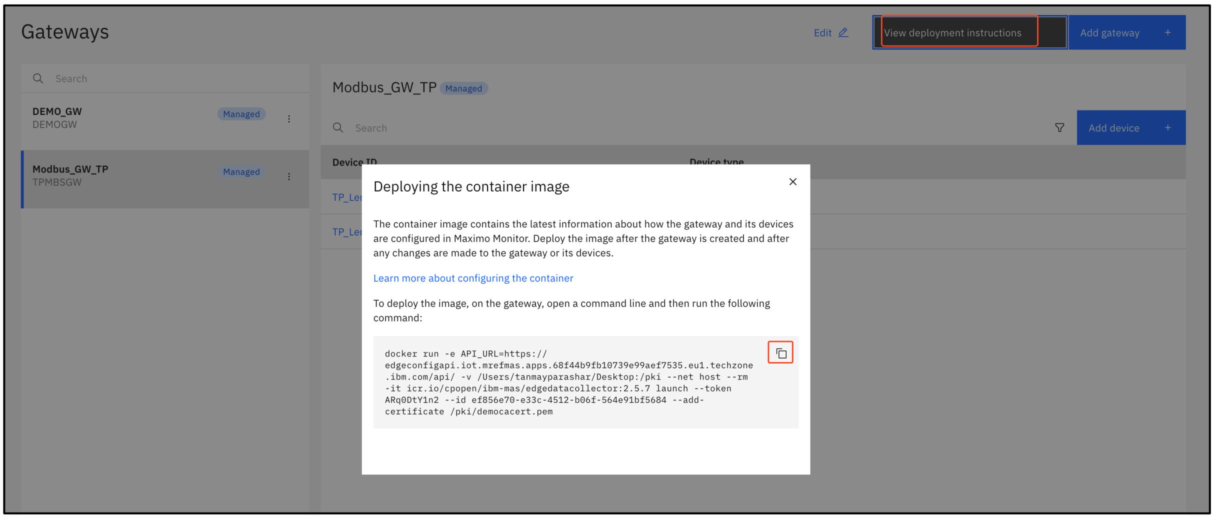
In this exercise the command would be: **docker kill 156c9fc21cb8**



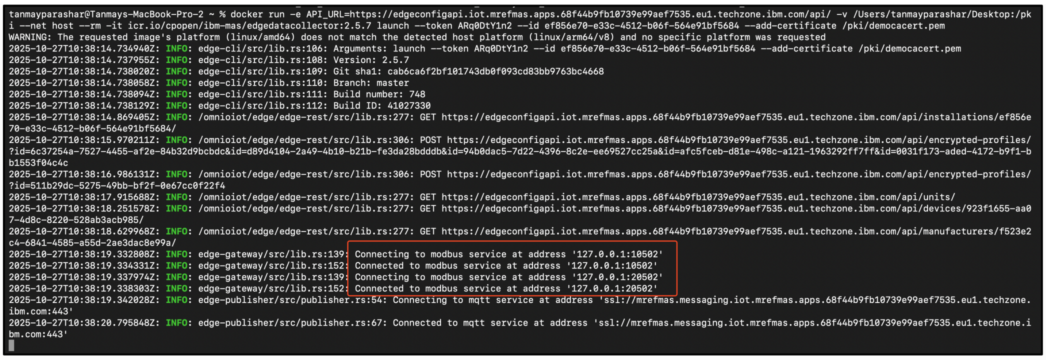
OR You can directly kill the container from Rancher Desktop Application by clicking on Stop.



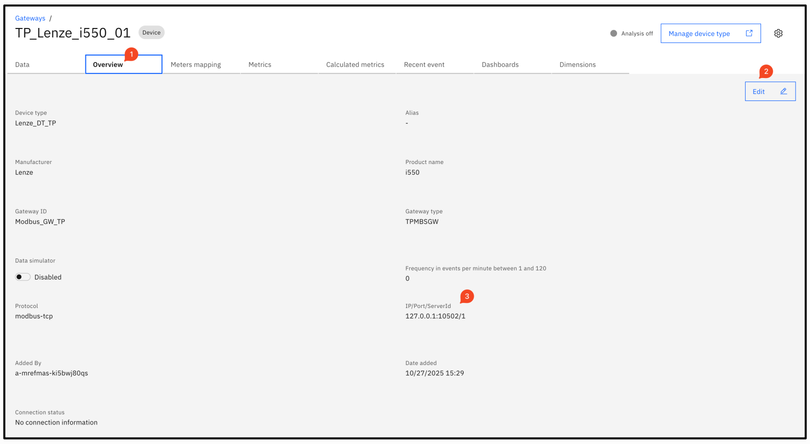
5.Navigate back to your Managed Gateway in Monitor and press the **View deployment instructions**. Click on the docker command to copy it to the clipboard.



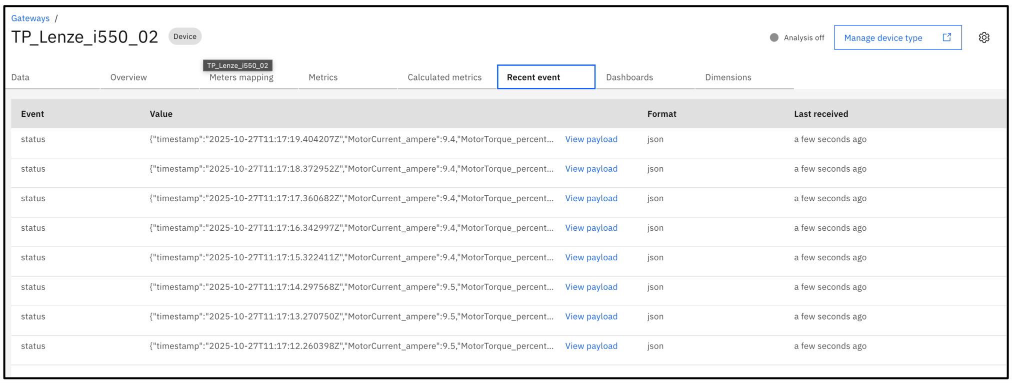
6. Get back to the terminal and then paste the docker command line from the clipboard. Click enter to execute it, and you should see something similar to the following.



NOTE: In this step, the IP address is set to localhost. This situation occurs when the network does not have a fixed IP address. If you notice that your network configuration or IP address (as seen using the **ifconfig en0** command) changes frequently, switch to your local IP address (commonly **127.0.0.1**, though it may vary depending on your system). Then, open the device you have created in your setup and update its IP address accordingly.

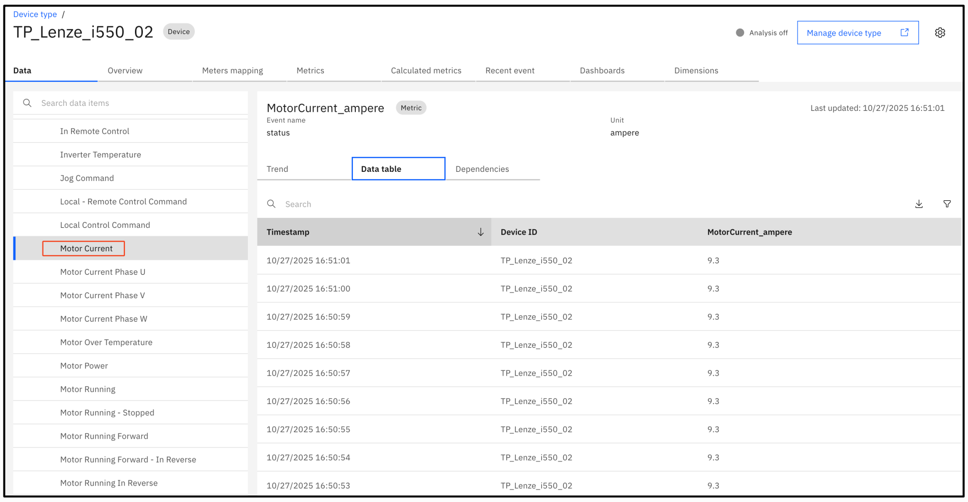


7. Navigate to **Recent event** for the second device and wait for a minute until the first message is coming through.

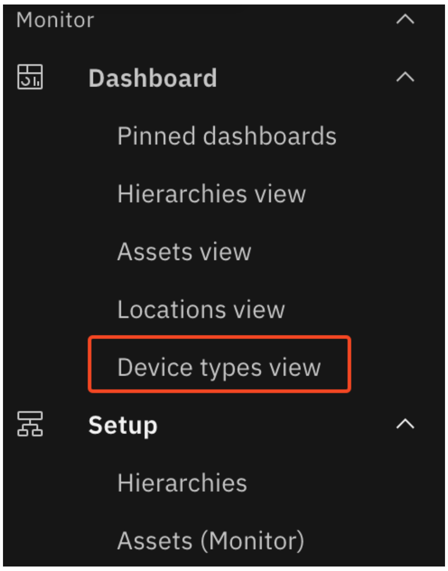


8. Click on **View payload** and see the data points being send to the Event name status.

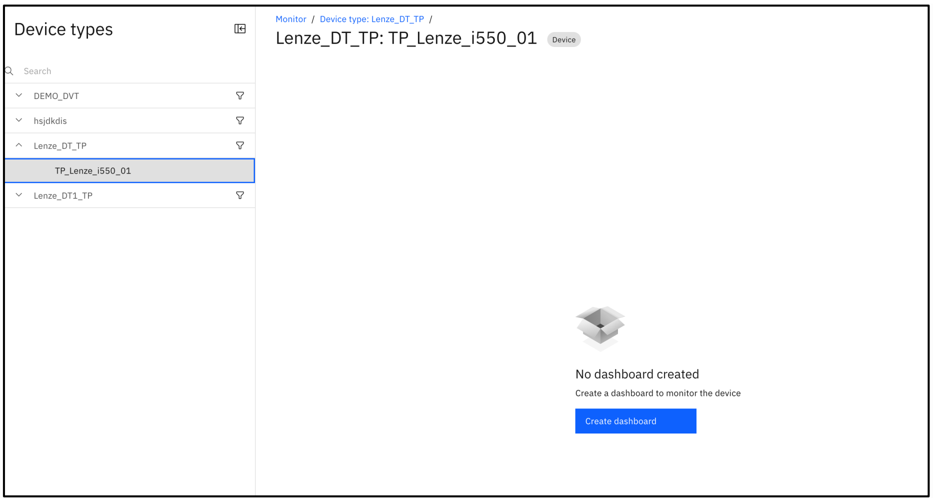
9. To see the data in the metrics. **Go to Data -> Motor Current -> Data Table.**



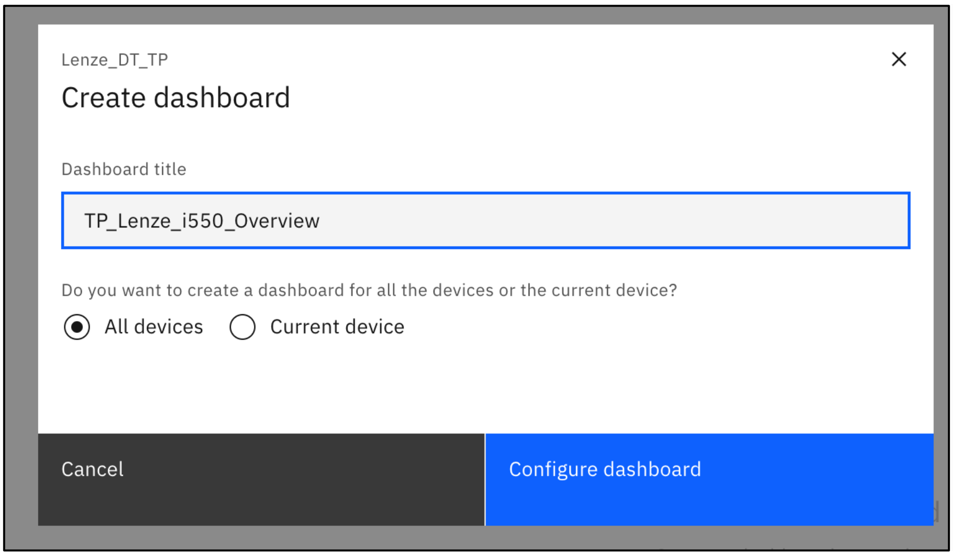
10. Navigate to **Dashboard -> Device Type View** in Monitor.



11. Select the **XX_Lenze_i550_01** device and then click on **Create Dashboard**

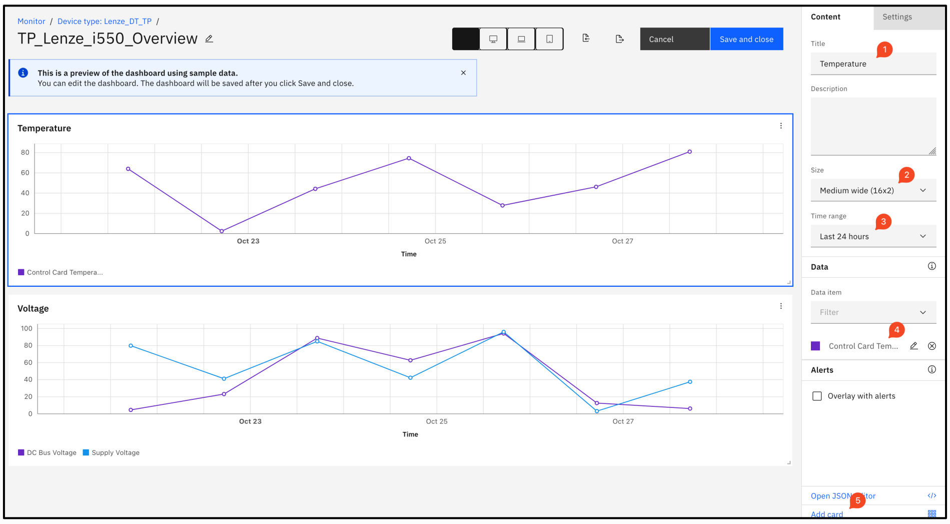


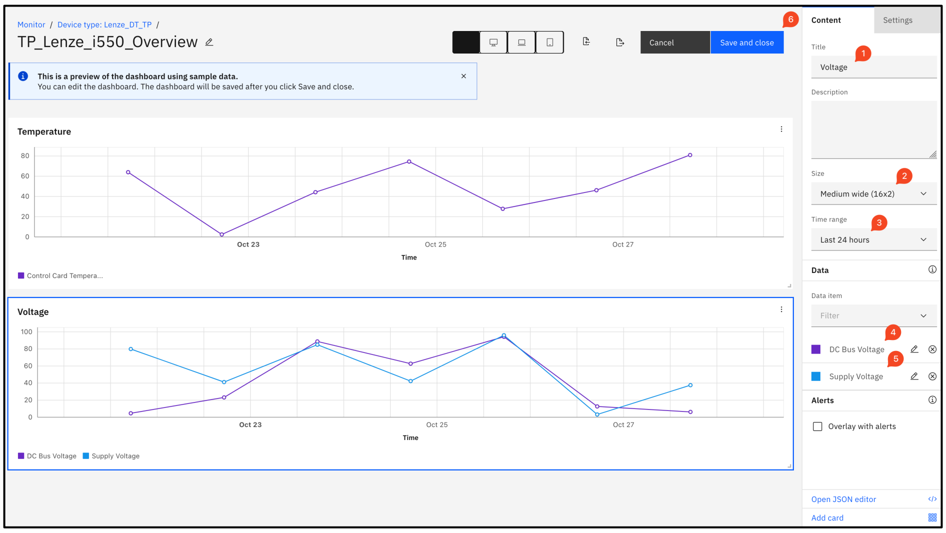
12. Give your new dashboard a name and press **Configure dashboard**.
    Title: **XX_Lenze_i550_Overview**



13. You will now see an empty dashboard configuration. Add a Time series line card by clicking on it.

- Locate the Control Card Temperature in the Data item dropdown and change the Time range to Last 24 hours.
- Give the card a title, like Temperature [°C] and drag the lower right corner to resize the card to full width.
- Once that is in place, then Add another Time series line card.
- Configure the second Time series line card according to the screen shot below and press Save and Close.





14. The dashboard for the first device will look like this.

15. Similarly you can configure for second device for different metrics.